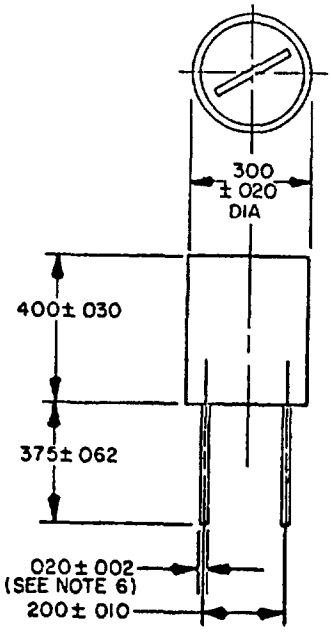
FED SUP CLASS
5950




| Inches | mm |
|---|---|
| .002 | 0 05 |
| 010 | 0 25 |
| .020 | 0 51 |
| .030 | 0 76 |
| .062 | 1.57 |
| .200 | 5 08 |
| .300 | 7.62 |
| .375 | 9.53 |
| 400 | 10 16 |

**Ratings**

| | |
|---|---|
| Style | LT10 |
| Grade | 1 |
| Class | A |
| Operating temperature range | -55°C to+105°C |
| Ambient temperature | 90°C max |
| Temperature rise | 15°C max |
| Working voltage | 300 Vdc |
| Percent coupling | 3 percent max |
| Dielectric withstanding voltage | |
| Sea level | 840 Vrms min |
| Reduced barometric pressure | 630 Vrms min |
| Terminal pull | 3 pounds min |
| Tuning torque | 0.40 to 6 in-oz. (D) |
| Weight | 1.5 grams max |
| Altitude | 70,000 FT. |

(D) DENOTES CHANGE

APPROVED 11 JAN 1973 REVISED (A) 7 NOV 1973 (B) 17 AUG 1979 (C) 4 FEB 80 (D) 8 MAY 87

User activities: Army - Navy - Air Force - 19

Review activities: Army - MI; Navy -; Air Force - 11, 17, 99; DLA - ES

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| | | | |
|---|---|---|---|
| PA Air Force 85<br>Other Cust<br>EC<br>ER | International Interest | TITLE COILS, RADIO FREQUENCY, MOLDED VARIABLE, SUBMINIATURE, IRON CORE, MAGNETICALLY SHIELDED, TYPES LT10V174 TO LT10V209 INCL AND LT10V247 TO LT10V258 INCL. | MILITARY STANDARD<br>MS21381 |
| Procurement Specification<br>MIL-C-15305 | | SUPERSEDES<br>MS21381(USAF) | PAGE 1 OF 5 |

DD FORM 672 (Continued) previous edition of this form are obsolete

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5950-0684-01

Electrical characteristics (initial)

| Dash No. 1/ 2/ | Type designation | Inductance tuning range | | | Test frequency (L&Q) | Q min at L nom | Minimum self resonant frequency at L nom | DC resistance (max) | Rated DC current (max) |
|---|---|---|---|---|---|---|---|---|---|
| | | L nom | L min | L max | | | | | |
| | | uH | uH | uH | MHz | | MHz | ohms | mA |
| -1 | LT10V174 | .10 | .095 | .105 | 25.0 | 45 | 250 | .030 | 1510 |
| -74 | LT10V247 | .11 | .105 | .116 | 25.0 | 45 | 250 | .030 | 1480 |
| -2 | LT10V175 | .12 | .114 | .126 | 25.0 | 45 | 250 | 030 | 1450 |
| -75 | LT10V248 | .13 | .124 | .137 | 25.0 | 45 | 250 | .030 | 1420 |
| -3 | LT10V176 | .15 | 143 | .158 | 25.0 | 45 | 250 | 030 | 1400 |
| -76 | LT10V249 | .16 | .152 | .168 | 25.0 | 45 | 250 | 035 | 1380 |
| -4 | LT10V177 | 18 | 171 | .189 | 25.0 | 45 | 250 | 035 | 1370 |
| -77 | LT10V250 | .20 | .190 | .210 | 25.0 | 45 | 250 | .035 | 1350 |
| -5 | LT10V178 | 22 | .209 | .231 | 25.0 | 45 | 250 | .035 | 1340 |
| -78 | LT10V251 | .24 | .228 | .252 | 25.0 | 45 | 250 | .040 | 1320 |
| -6 | LT10V179 | .27 | .257 | .284 | 25.0 | 51 | 250 | 040 | 1300 |
| -79 | LT10V252 | .30 | .285 | .315 | 25.0 | 51 | 250 | .040 | 1280 |
| -7 | LT10V180 | .33 | .314 | .347 | 25.0 | 51 | 250 | .040 | 1260 |
| -80 | LT10V253 | .36 | .342 | .378 | 25.0 | 51 | 250 | 045 | 1250 |
| -8 | LT10V181 | .39 | .371 | .410 | 25.0 | 51 | 210 | 045 | 1240 |
| -81 | LT10V254 | .43 | .409 | 452 | 25.0 | 51 | 200 | .045 | 1220 |
| -9 | LT10V182 | 47 | .447 | .494 | 25 0 | 51 | 184 | .045 | 1200 |
| -82 | LT10V255 | .51 | .485 | .536 | 25 0 | 51 | 180 | 050 | 1180 |
| -10 | LT10V183 | .56 | .532 | .588 | 25.0 | 51 | 176 | .050 | 1160 |
| -83 | LT10V256 | .62 | .589 | .651 | 25.0 | 51 | 163 | 055 | 1130 |
| -11 | LT10V184 | .68 | .646 | .714 | 25.0 | 51 | 152 | .055 | 1100 |
| -84 | LT10V257 | .75 | .713 | .788 | 25.0 | 51 | 147 | .060 | 1070 |
| -12 | LT10V185 | .82 | .779 | .861 | 25.0 | 55 | 144 | .060 | 1040 |
| -85 | LT10V258 | .91 | .865 | .956 | 25.0 | 55 | 136 | .070 | 1010 |
| -13 | LT10V186 | 1.00 | .950 | 1.050 | 25.0 | 55 | 128 | .070 | 986 |
| -14 | LT10V187 | 1.2 | 1.08 | 1.32 | 7.9 | 58 | 136 | .085 | 968 |
| -15 | LT10V188 | 1.5 | 1.35 | 1.65 | 7.9 | 64 | 124 | 100 | 893 |
| -16 | LT10V189 | 1.8 | 1.62 | 1.98 | 7.9 | 74 | 108 | .110 | 853 |
| -17 | LT10V190 | 2.2 | 1.98 | 2 42 | 7.9 | 71 | 96 | 120 | 817 |
| -18 | LT10V191 | 2.7 | 2 43 | 2.97 | 7.9 | 71 | 83.2 | 125 | 800 |
| -19 | LT10V192 | 3.3 | 2.97 | 3.63 | 7.9 | 58 | 74.4 | .165 | 696 |
| -20 | LT10V193 | 3.9 | 3.51 | 4.29 | 7.9 | 58 | 69.6 | .180 | 659 |

User activities: Army - ; Navy - ; Air Force - 19
Review activities: Army - MI; Navy - ; Air Force -11, 17, 99; DLA - ES
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s military standard is approved for use by all Departments and Agencies of the
partment of Defense Selection for all new engineering and design applications
l for repetitive use shall be made from this document when applicable

Electrical characteristics (initial) Continued

| Dash No 1/ 2/ | Type designation | Inductance tuning range | | | Test Frequency (L&Q) | Q min at L nom | Minimum self resonant frequency at L nom | DC resistance (max) | Rated DC current (max) |
|---|---|---|---|---|---|---|---|---|---|
| | | L nom | L min | L max | | | | | |
| | | uH | uH | uH | MHz | | MHz | ohms | mA |
| -21 | LT10V194 | 4 7 | 4 23 | 5.17 | 7.9 | 61 | 63.2 | 245 | 571 |
| -22 | LT10V195 | 5.6 | 5 04 | 6.16 | 7.9 | 61 | 57 6 | 265 | 550 |
| -23 | LT10V196 | 6.8 | 6.12 | 7.46 | 7 9 | 55 | 50 4 | .330 | 493 |
| -24 | LT10V197 | 8.2 | 7 38 | 9.02 | 7.9 | 61 | 48 0 | .460 | 417 |
| -25 | LT10V198 | 10 | 9 00 | 11 0 | 7 9 | 58 | 43 2 | .640 | 359 |
| -26 | LT10V199 | 12 | 10 8 | 13.2 | 2 5 | 77 | 29 6 | 800 | 316 |
| -27 | LT10V200 | 15 | 13 5 | 16.5 | 2.5 | 77 | 23 0 | 865 | 301 |
| -28 | LT10V201 | 18 | 16 2 | 19.8 | 2 5 | 74 | 19 0 | .940 | 292 |
| -29 | LT10V202 | 22 | 19 8 | 24 2 | 2.5 | 80 | 17.0 | 1 03 | 267 |
| -30 | LT10V203 | 27 | 24.3 | 29 7 | 2.5 | 74 | 16.5 | 1 18 | 243 |
| -31 | LT10V204 | 33 | 29.7 | 36 3 | 2.5 | 77 | 14 9 | 1 30 | 231 |
| -32 | LT10V205 | 39 | 35.1 | 42.9 | 2 5 | 77 | 14 1 | 1 41 | 223 |
| -33 | LT10V206 | 47 | 42.3 | 51.7 | 2.5 | 71 | 11.9 | 1 61 | 203 |
| -34 | LT10V207 | 56 | 50.4 | 61.6 | 2.5 | 74 | 11.1 | 2.08 | 191 |
| -35 | LT10V208 | 68 | 61 2 | 74 8 | 2.5 | 67 | 10.3 | 2 20 | 185 |
| -36 | LT10V209 | 82 | 73.8 | 90 2 | 2 5 | 67 | 9.35 | 2 42 | 174 |

1/ The dash number added to the MS military standard number constitutes the MS part number, for example MS21381-1.
2/ See note 9

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User activities Army -
Navy -
Air Force - 19

Review activities Army - MI
Navy -
Air Force - 11, 17, 99
DLA - ES

Electrical Characteristics (Final)

| Inspection Group | Allowable variation from initial measurement | | Allowable % from specified minimum value in Electrical Characteristics (initial) table. | |
|---|---|---|---|---|
| | 1/ Inductance Percent | DC resistance | Self-resonant frequency | Q |
| Qualification Inspection | | | | |
| Group II | ±2 | ---- | ---- | -10 |
| Group III | ±5 | ±(3%+.001 ohm) | -8 | -10 |
| Group IV | ±5 | ±(2% +.001 ohm) | -10 | -10 |
| Quality Conformance Inspection | | | | |
| Group C | | | | |
| Subgroup I | ±2 | ---- | ---- | -10 |
| Subgroup II | ±5 | ±(2% +.001 ohm) | -10 | -10 |
| Subgroup III | ±5 | ±(3% +.001 ohm) | -8 | -10 |

1/ Initial inductance shall consist of testing the variable for the inductance range; equal to or less than MIN.L and equal to or more than MAX. L.

The variable shall then be set to the nominal inductance.

The remaining applicable electrical characteristics shall be read without readjusting the unit.

Allowable variation from initial inductance shall be the precent change between nominal inductance and the final inductance reading.

The variable shall not be reset or adjusted between initial and final inductance tests.

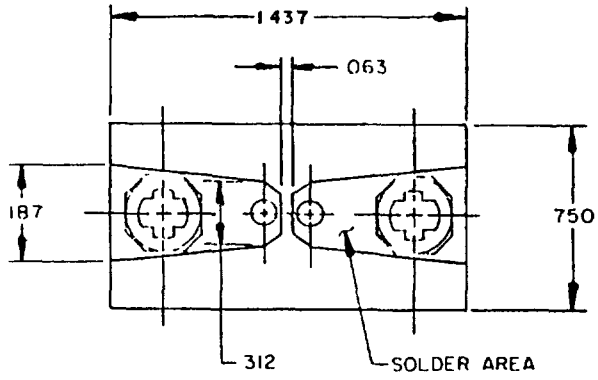
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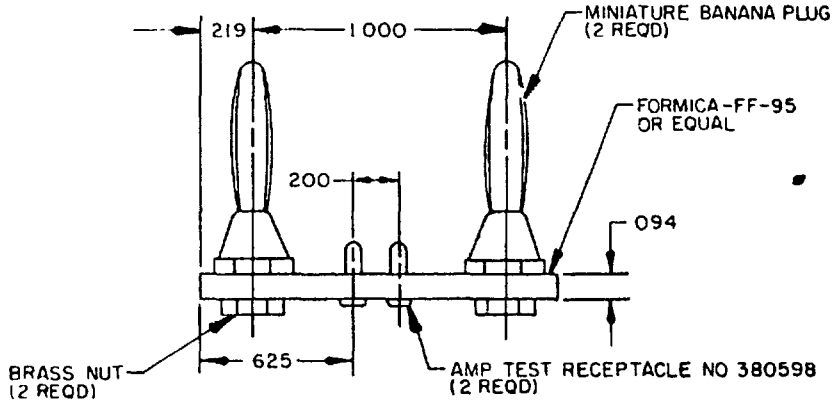
| P.A | International interest | TITLE | MILITARY STANDARD |
|---|---|---|---|
| Air Force - 85<br>Other Cust<br>EC<br>ER | | COILS, RADIO FREQUENCY, MOLDED VARIABLE, SUBMINIATURE, IRON CORE, MAGNETICALLY SHIELDED, TYPES LT10V174 TO LT10V209 INCL. AND LT10V247 TO LT10V258 INCL. | MS 21381 |
| Procurement Specification<br>MIL-C-15305 | | SUPERSEDES:<br>MS21381(USAF) | PAGE 4 OF 5 |



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| INCH | mm |
|---|---|
| 063 | 1.60 |
| 094 | 2 39 |
| 187 | 4 75 |
| 200 | 5 08 |
| 219 | 5 56 |
| 312 | 7 92 |
| 625 | 15 88 |
| 750 | 19.05 |
| 1 000 | 25 40 |
| 1 437 | 36 50 |




## TEST FIXTURE FOR ELECTRICAL MEASUREMENTS

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These coils are intended to be mounted by the body.
4. Barometric pressure test (test condition C) is applicable.
5. Shock, specified pulse, method 213, test condition I, is applicable.
6. Tinned copper lead wire, AWG No. 24.
7. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
8. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this standard to the extent specified herein.
9. Former MS part numbers MS21381-37 thru MS21381-73 have been superseded by MS21402-1 thru MS21402-37, respectively.
10. Tolerance is ±.005 (0.13 mm).

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User activities: Army - Navy - Air Force - 19
Review activities: Army - MI Navy - Air Force - 11, 17, 99 DLA - ES

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| | | | |
|---|---|---|---|
| P A<br>Air Force - 85<br>Other Cust<br>EC<br>ER | International Interest | TITLE COILS, RADIO FREQUENCY, MOLDED VARIABLE, SUBMINIATURE, IRON CORE, MAGNETICALLY SHIELDED, TYPES LT10V174 TO LT10V209 INCL. AND LT10V247 TO LT10V258 INCL | MILITARY STANDARD<br>MS21381 |
| Procurement Specification<br>MIL-C-15305 | | SUPERSEDES<br>MS21381(USAF) | PAGE 5 OF 5 |



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